Definizione

Sia $f : I \setminus \{x_0\} \to \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$

f AMMETTE UNA PROPRIETÀ DEFINITIVAMENTE PER $x \to x_0$ ("PER x CHE TENDE A x_0")

SE $\exists\, U$ intorno di x_0 tale che

$\forall x \in I \cap U$, $x \neq x_0$, $f(x)$ VERIFICA QUELLA PROPRIETÀ

ES: $f(x) = \dfrac{1}{x^2 - 1}$ f è positiva per $x \to 2$

DEFINIZIONE TOPOLOGICA DI LIMITE

Siano $x_0 \in \overline{\mathbb{R}}$, $\ell \in \overline{\mathbb{R}}$

Sia $f : I \setminus \{x_0\} \to \mathbb{R}$

Si dice che

$$\lim_{x \to x_0} f(x) = \ell$$

se e solo se

$\forall V$ intorno di ℓ risulta che

$f(x) \in V$ DEFINITIVAMENTE per $x \to x_0$

(se x è "vicino" a x_0, allora

$f(x)$ è "vicino" a ℓ)

Allora (più esplicitamente)

$\lim_{x \to x_0} f(x) = l$ se e solo se

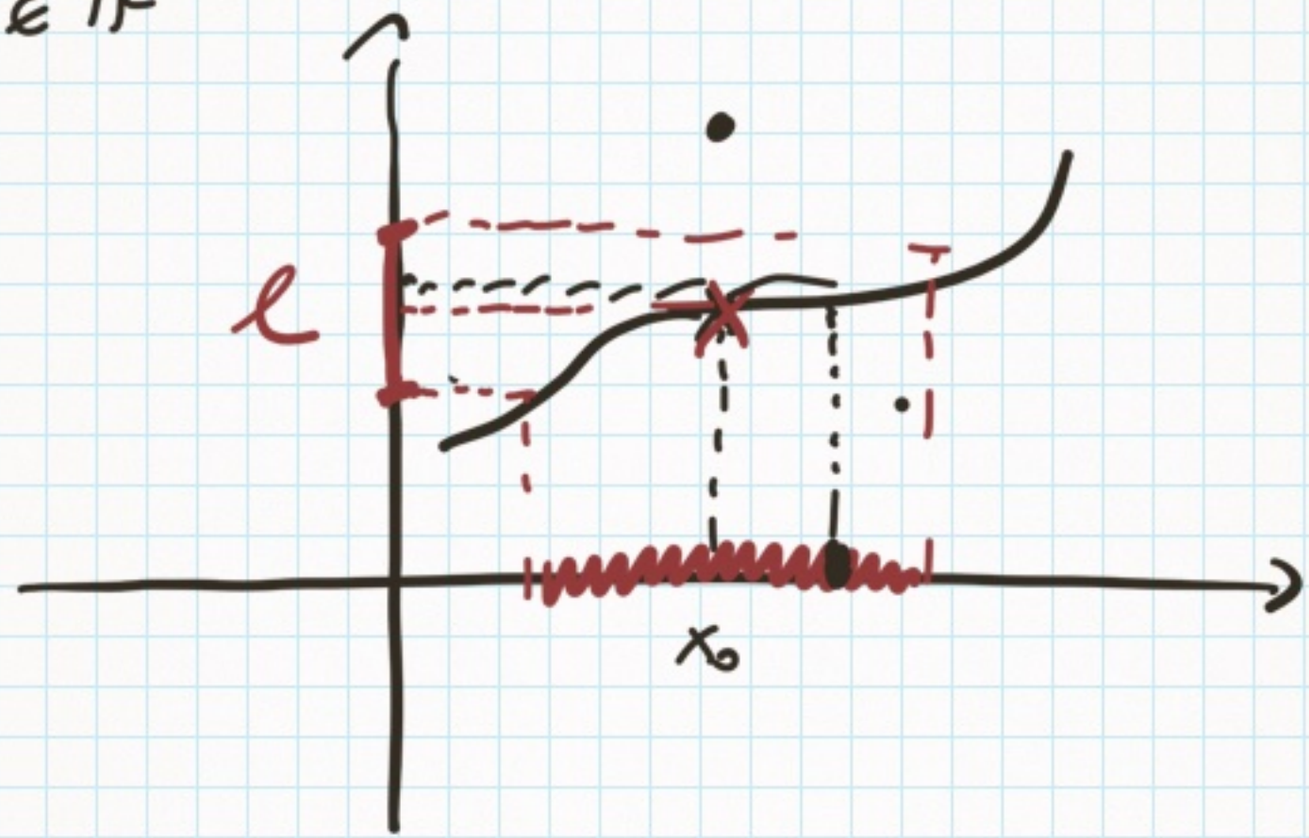
$\forall$ V intorno di l $\exists$ U intorno di x_0 tale che $\forall x \in I \cap U$, $x \neq x_0$,

$f(x) \in V$

↓

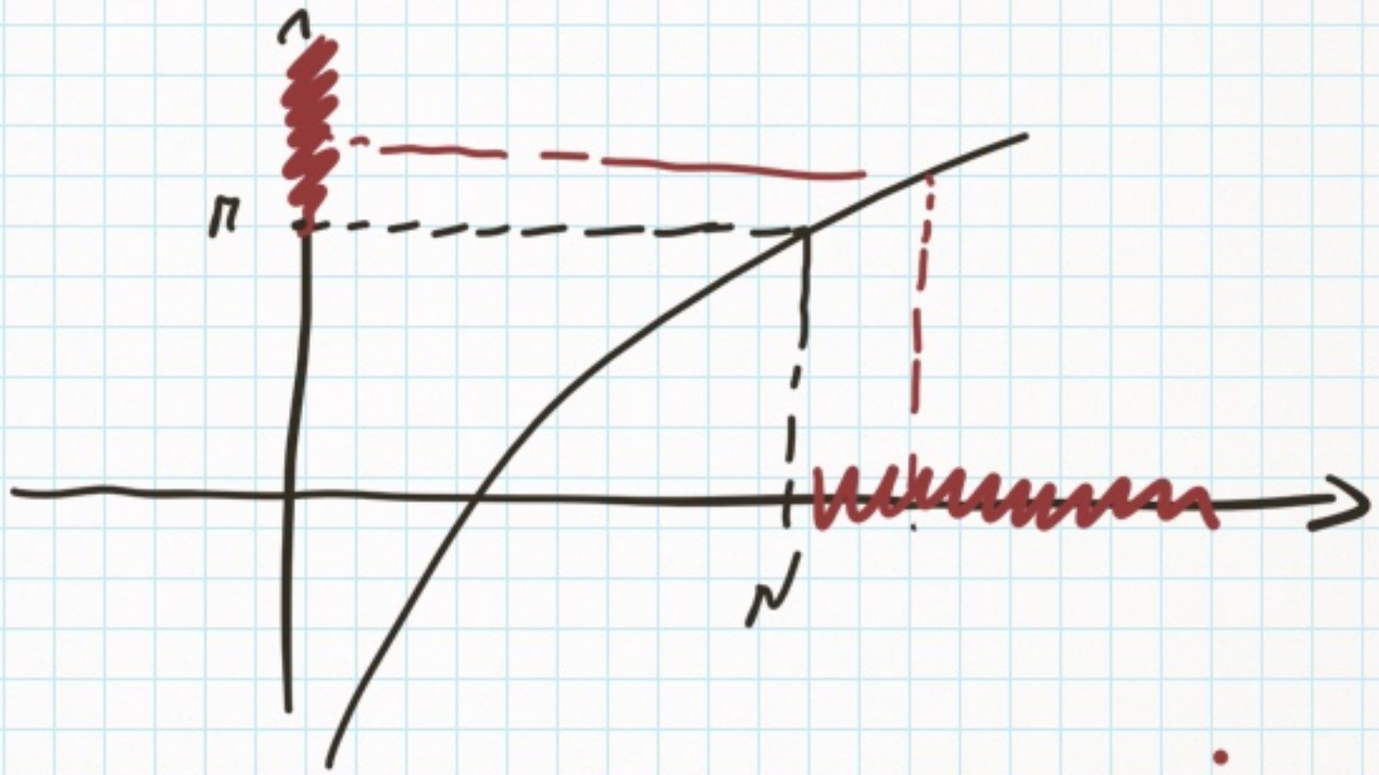
DEFINITIVAMENTE per $x \to x_0$.

$l, x_0 \in \mathbb{R}$



$x_0, l = +\infty$

$\lim_{x \to +\infty} f(x) = +\infty$



Sia $x_0 \in \mathbb{R}$

Diremo che "limite per x che tende a x_0 DA DESTRA"

$\lim_{x \to x_0^+} f(x) = l \iff \forall V$ intorno di l

$f(x) \in V$ definitivamente per $x \to x_0^+$

(cioè "si avvicina a x_0" DA DESTRA")

ossia

$\exists U$ intorno destro di x_0 t.c.

$\forall x \in U \cap I,\ x \neq x_0,\quad f(x) \in V$

Diremo che " x che tende a x_0 da sinistra "

$\lim_{x \to x_0^-} f(x) = l \iff \forall$ V intorno di l

$f(x) \in V$ definitivamente per $x \to x_0^-$

ovvero

$\forall$ V intorno di l $\exists$ U intorno sinistro di x_0 tale che $\forall x \in U \cap I$, $x \neq x_0$

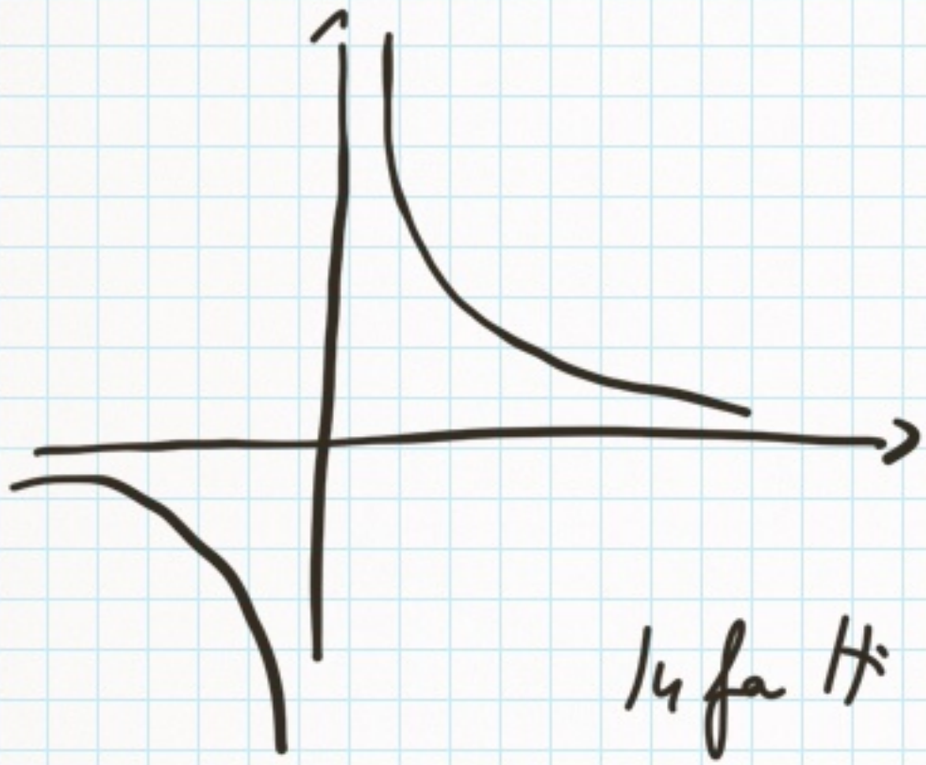
$f(x) \in V$

PROP.

$\exists \lim_{x \to x_0} f(x) = l \iff \lim_{x \to x_0^+} f(x) = l = \lim_{x \to x_0^-} f(x)$

ESEMPIO

Non esiste $\lim_{x \to 0} \frac{1}{x}$



Infatti

$\lim_{x \to 0^+} \frac{1}{x} = +\infty$ $\lim_{x \to 0^-} \frac{1}{x} = -\infty$

DEF. Sia $f : I \to \mathbb{R}$, I intervallo, $x_0 \in I$

f è CONTINUA in x_0 se e solo se

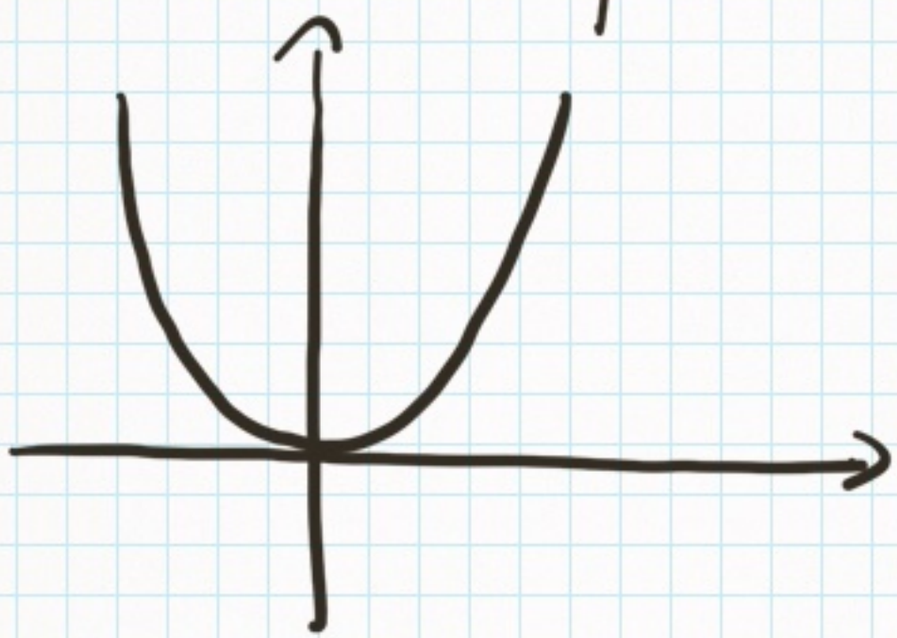
$$\lim_{x \to x_0} f(x) = f(x_0)$$

Una funzione è CONTINUA se è continua $\forall x_0 \in I$.

TUTTE LE FUNZIONI ELEMENTARI SONO CONTINUE

LIMITI DELLE FUNZIONI ELEMENTARI

(1) $f(x) = x^n$ n pari



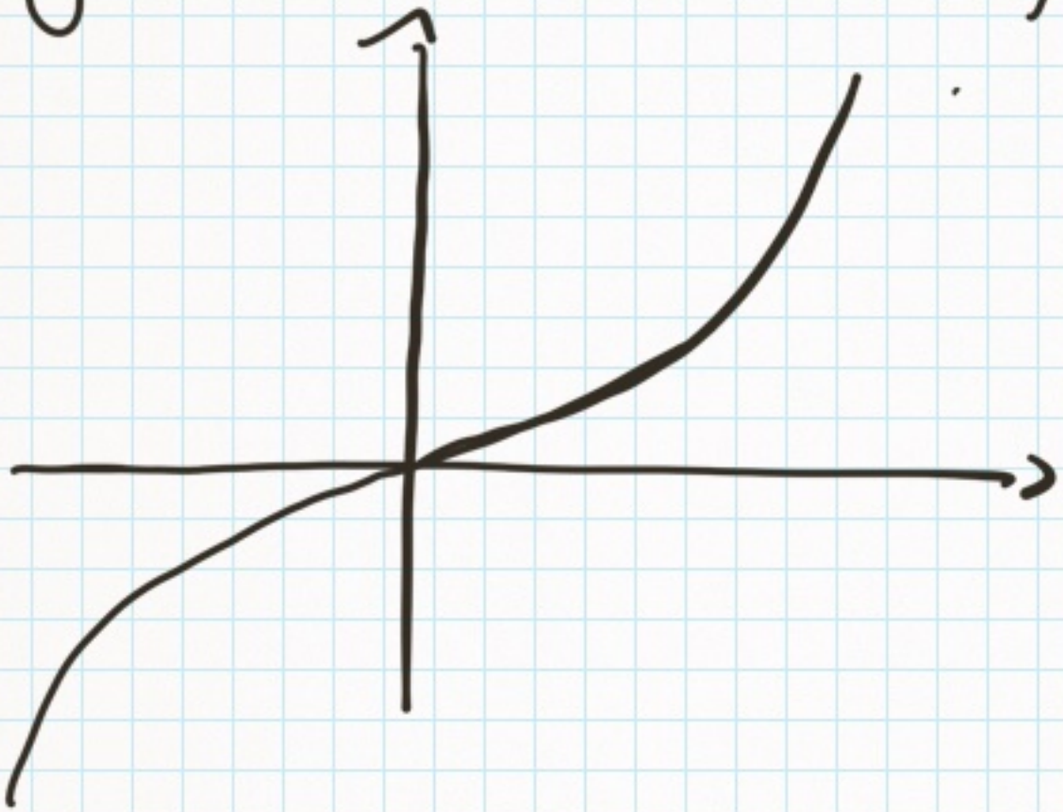
$$\lim_{x \to +\infty} x^n = +\infty$$

$$\lim_{x \to -\infty} x^n = +\infty$$

$\forall x_0 \in \mathbb{R}$

$$\lim_{x \to x_0} x^n = x_0^n$$

12. $f(x) = x^n$ n dispari



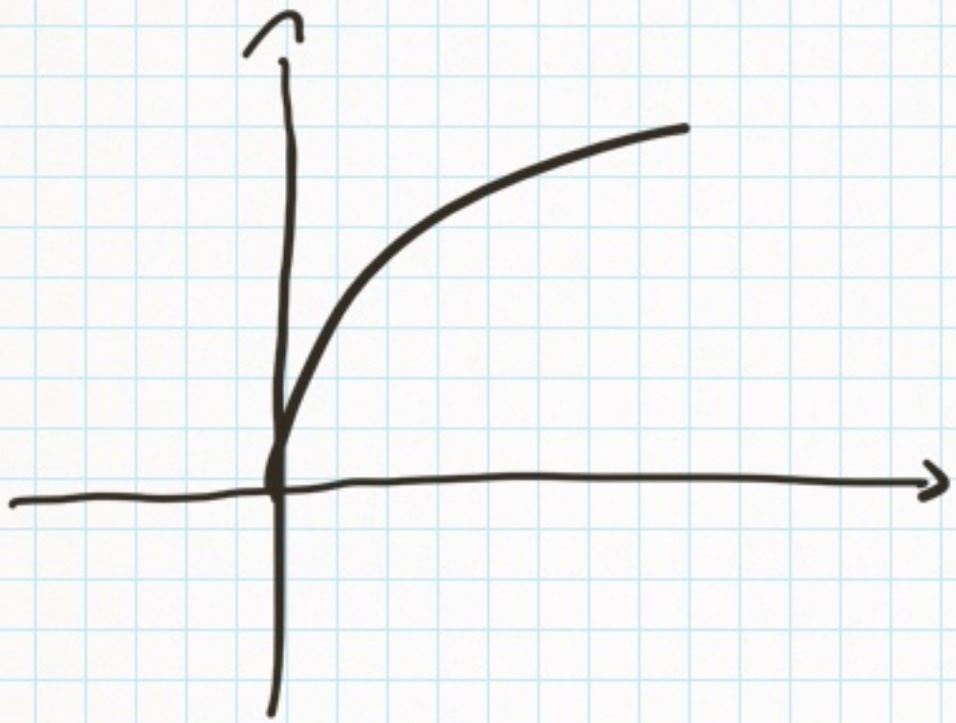
$$\lim_{x \to +\infty} x^n = +\infty$$

$$\lim_{x \to -\infty} x^n = -\infty$$

$\forall x_0 \in \mathbb{R}$

$$\lim_{x \to x_0} x^n = x_0^n$$

13) $f(x) = \sqrt[n]{x}$ n pari $x \geq 0$

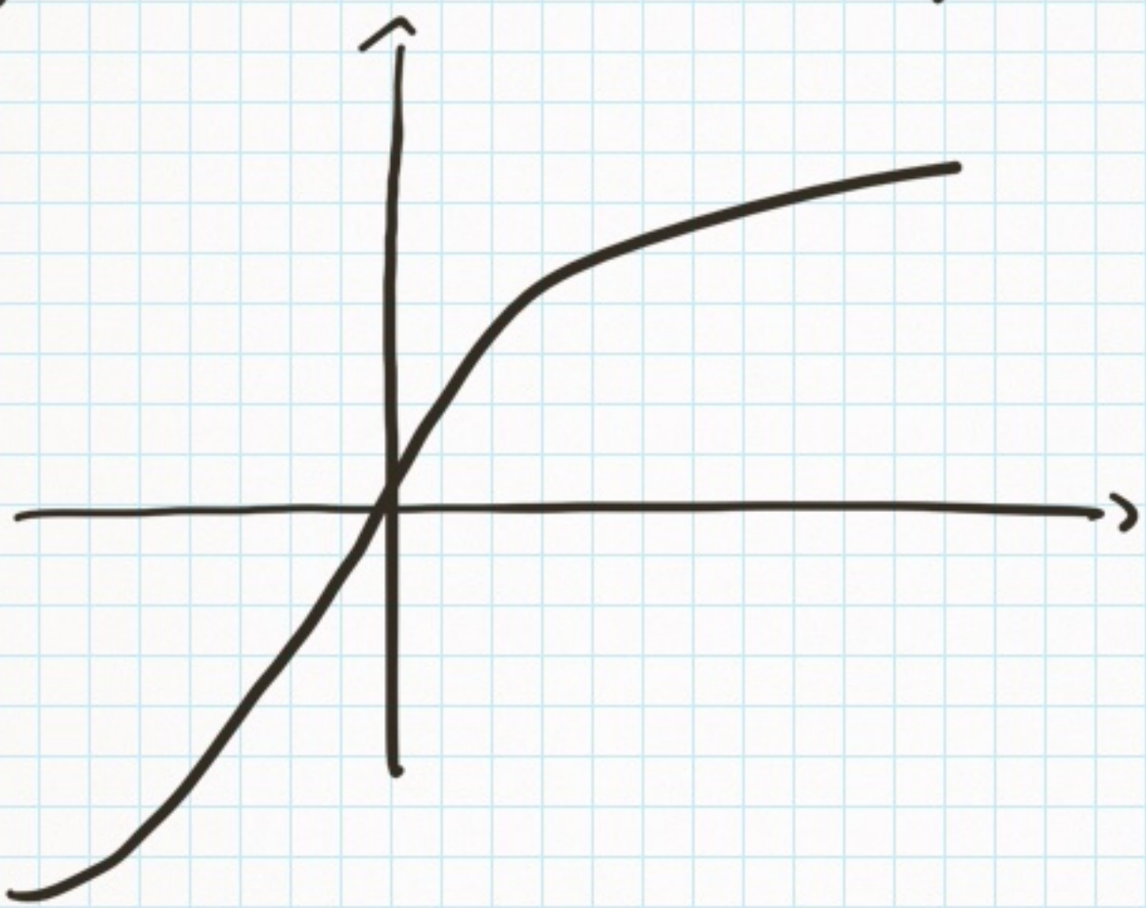


$$\lim_{x \to +\infty} \sqrt[n]{x} = +\infty$$

$\forall x_0 > 0$ $\lim_{x \to x_0} \sqrt[n]{x} = \sqrt[n]{x_0}$

$$\lim_{x \to 0^+} \sqrt[n]{x} = 0$$

(4) $f(x) = \sqrt[n]{x}$ n dispari



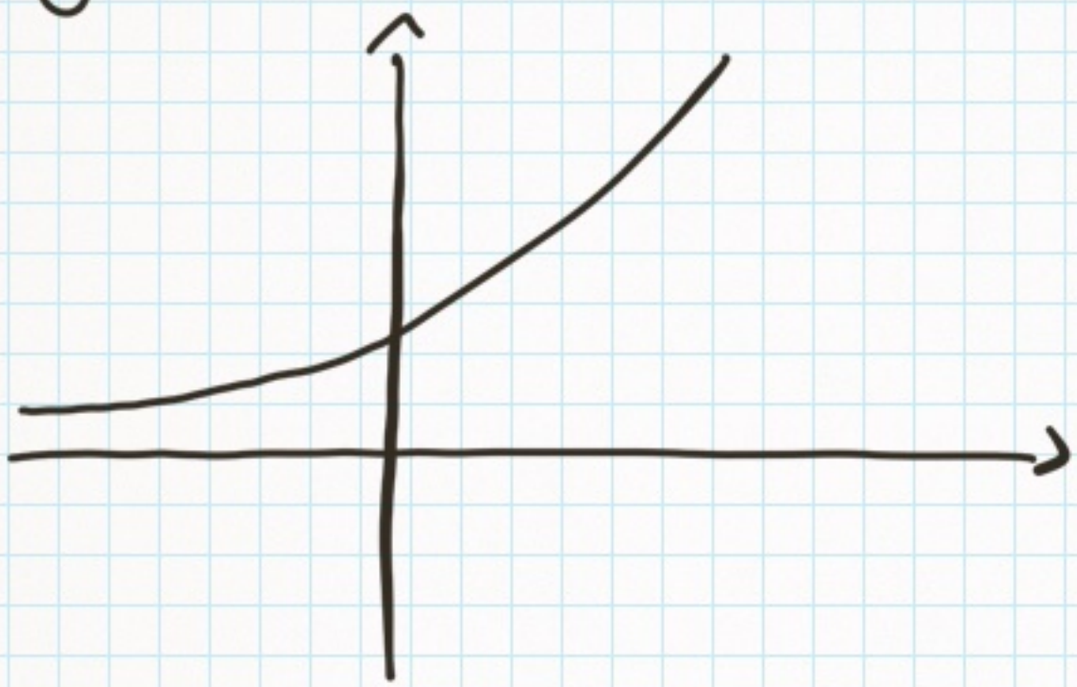
$\lim_{x \to -\infty} \sqrt[n]{x} = -\infty$

$\lim_{x \to +\infty} \sqrt[n]{x} = +\infty$

$\forall x_0 \in \mathbb{R}$

$\lim_{x \to x_0} \sqrt[n]{x} = \sqrt[n]{x_0}$

(5) $f(x) = a^x$, $a > 1$



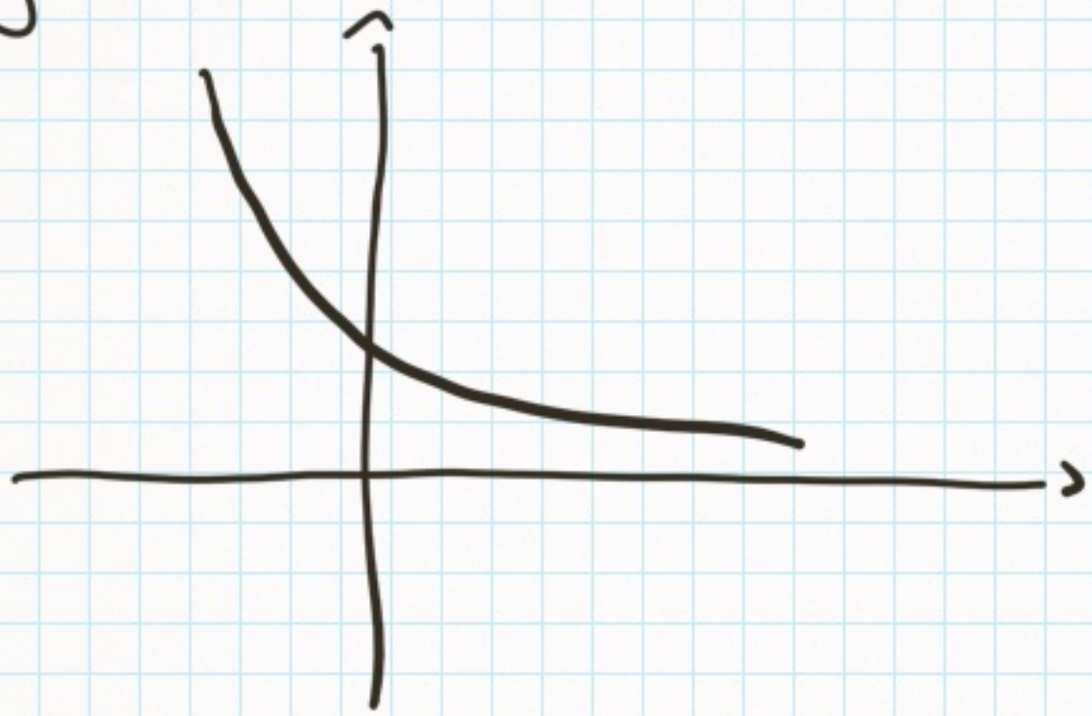
$$\lim_{x \to +\infty} a^x = +\infty$$

$$\lim_{x \to -\infty} a^x = 0$$

$\forall x_0 \in \mathbb{R}$

$$\lim_{x \to x_0} a^x = a^{x_0}$$

16) $f(x) = a^x$ $\quad 0 < a < 1$



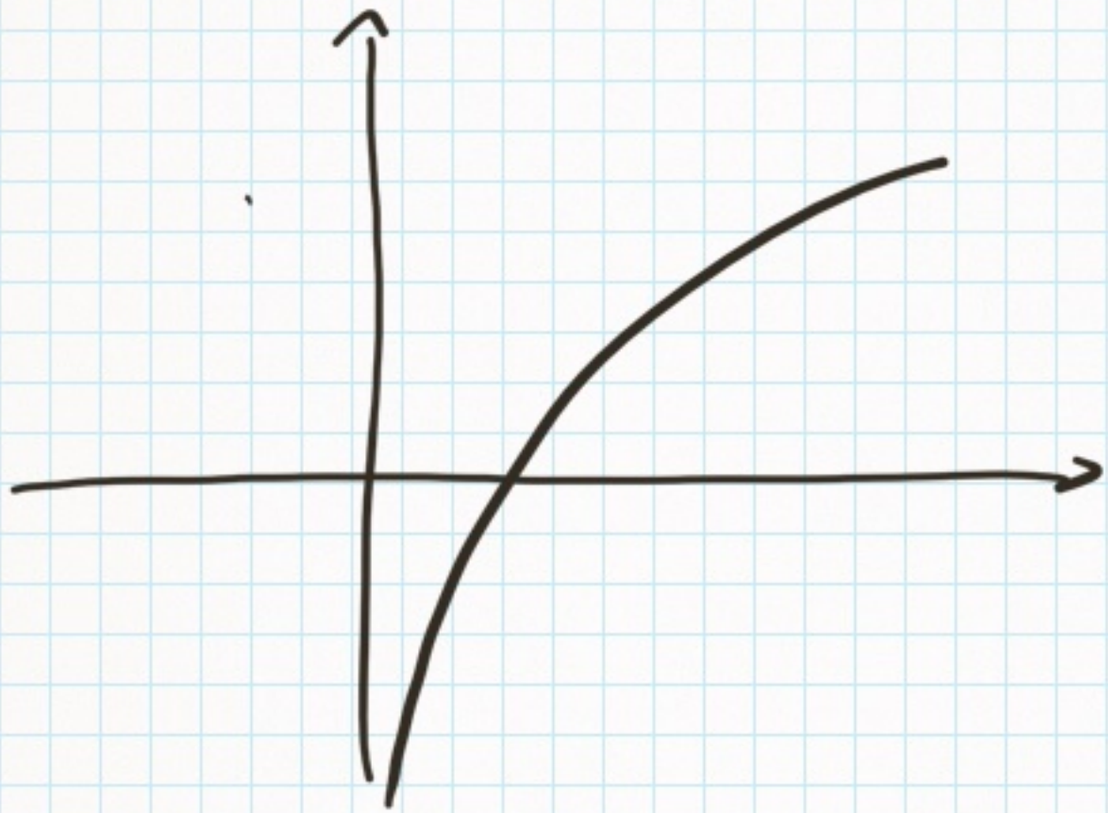
$$\lim_{x \to -\infty} a^x = +\infty$$

$$\lim_{x \to +\infty} a^x = 0$$

$\forall x_0 \in \mathbb{R} \quad \lim_{x \to x_0} a^x = a^{x_0}$

(7)

$$f(x) = \log_a x \qquad a > 1$$

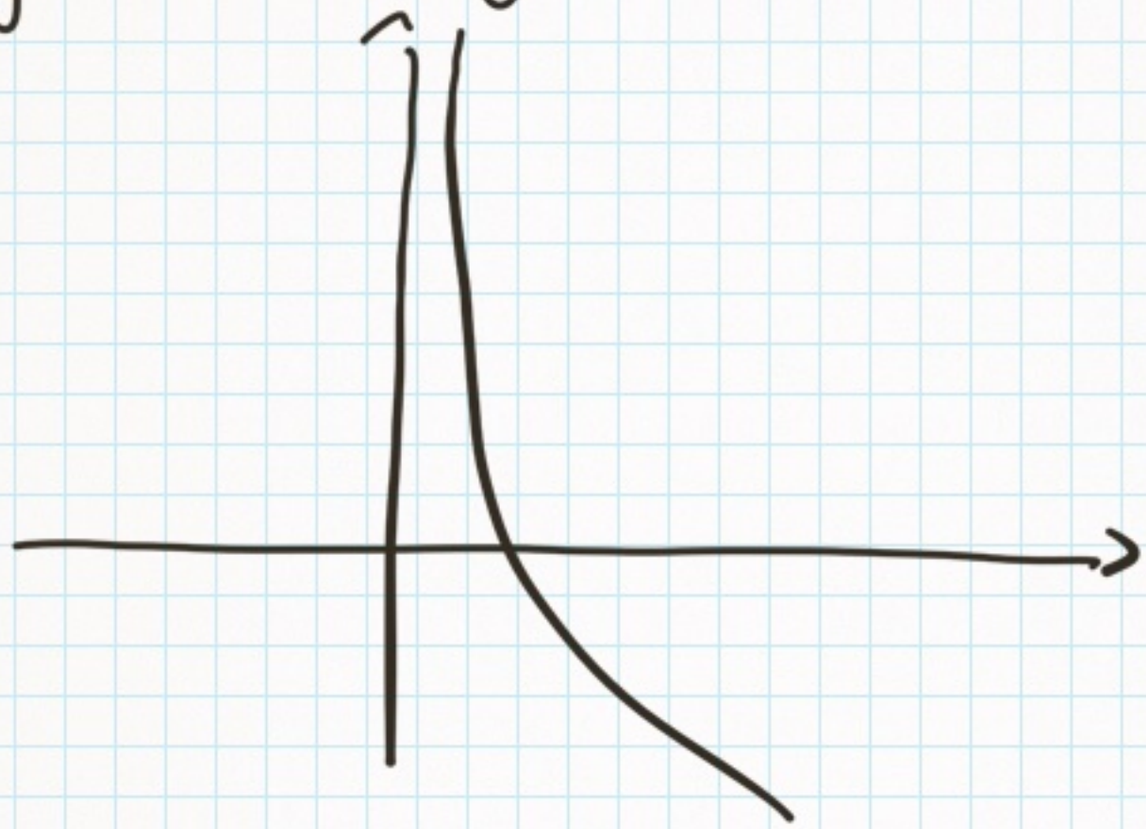


$$\lim_{x \to +\infty} \log_a x = +\infty$$

$$\lim_{x \to 0^+} \log_a x = -\infty$$

$$\forall x_0 > 0 \qquad \lim_{x \to x_0} \log_a x = \log_a x_0$$

(9) $f(x) = \log_a x$ $\quad 0 < a < 1$

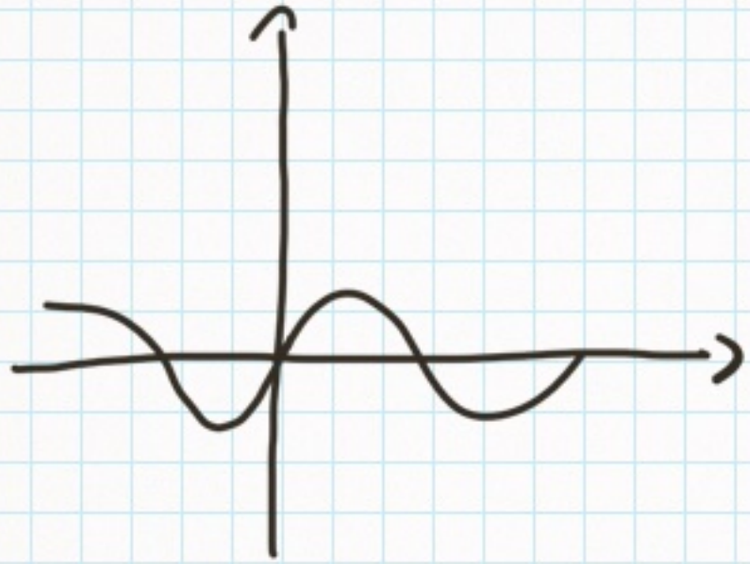


$$\lim_{x \to +\infty} \log_a x = -\infty$$

$$\lim_{x \to 0^+} \log_a x = +\infty$$

$$\forall x_0 > 0 \quad \lim_{x \to x_0} \log_a x = \log_a x_0$$

(9) $f(x) = \sin x$



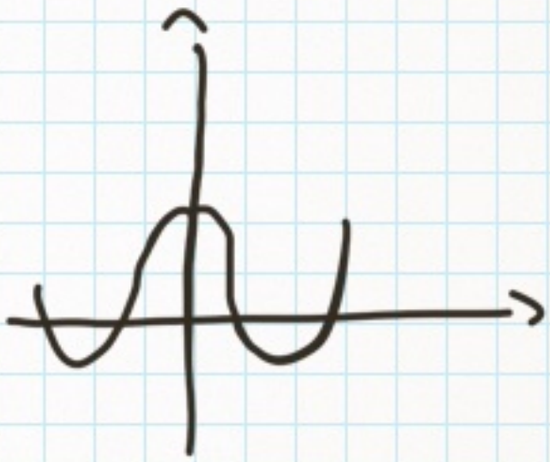
$\nexists \lim_{x \to \pm\infty} \sin x$

$\forall x_0 \in \mathbb{R} \quad \lim_{x \to x_0} \sin x = \sin x_0$

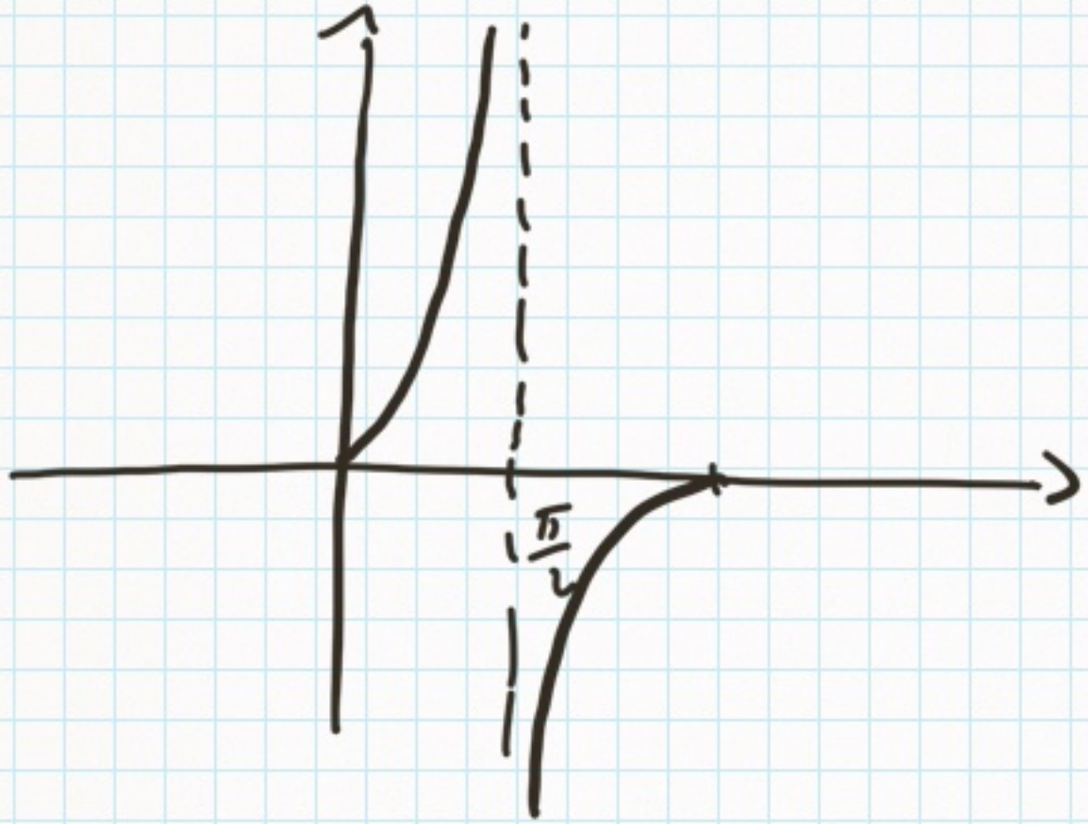
(10) $f(x) = \cos x$

$\nexists \lim_{x \to \pm\infty} \cos x$

$\forall x_0 \in \mathbb{R} \quad \lim_{x \to x_0} \cos x = \cos x_0$



(iii) $f(x) = \tan x$ $\quad x \neq \frac{\pi}{2}$



Vediamo per $x \in [0, \pi]$ (è una funzione periodica di periodo π)

$\lim_{x \to \frac{\pi}{2}^-} \tan x = +\infty$ $\qquad$ $\lim_{x \to \frac{\pi}{2}^+} \tan x = -\infty$

$\forall x \in [0, \pi]$, $x \neq \frac{\pi}{2}$ $\qquad$ $\lim_{x \to x_0} \tan x = \tan x_0$

I risultati si estendono per periodicità su tutto $\mathbb{R}$

e $\nexists \lim_{x \to \pm\infty} \tan x$

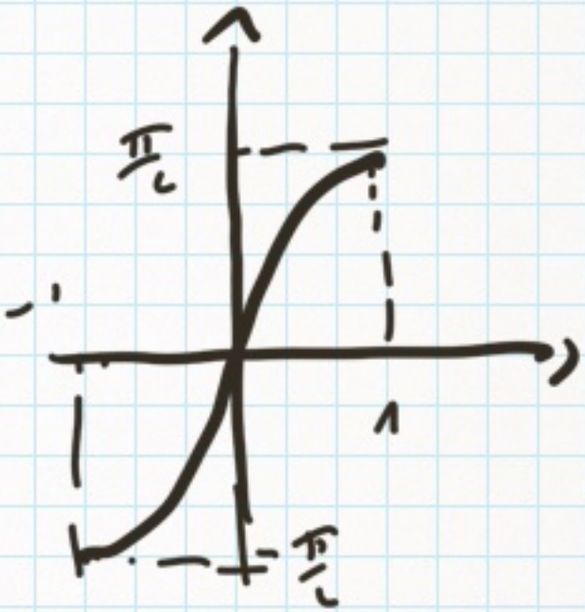
(12) $f(x) = \arcsin x \qquad -1 \le x \le 1$

$\forall \; x_0 \in]-1, 1[$

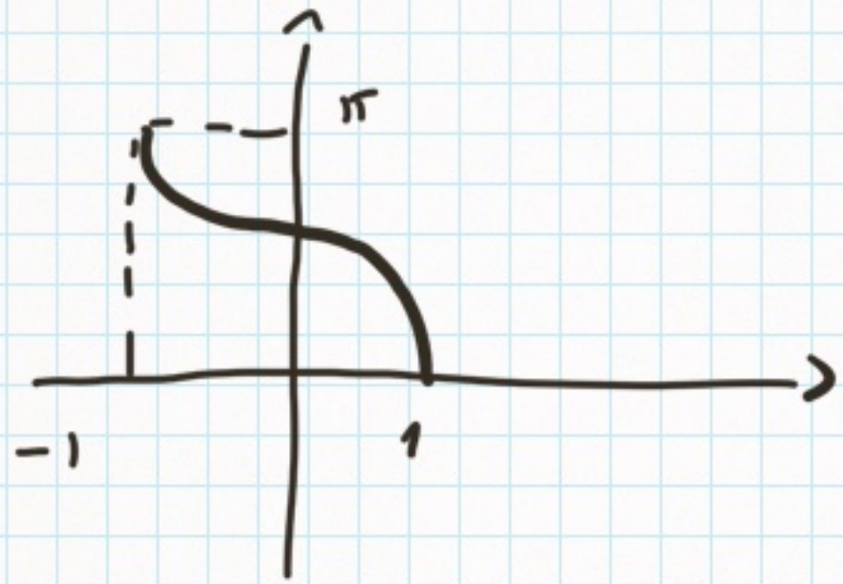
$\lim_{x \to x_0} \arcsin x = \arcsin x_0$

$\lim_{x \to -1^+} \arcsin x = -\frac{\pi}{2}$

$\lim_{x \to 1^-} \arcsin x = \frac{\pi}{2}$



(13) $f(x) = \arccos x \qquad -1 \le x \le 1$

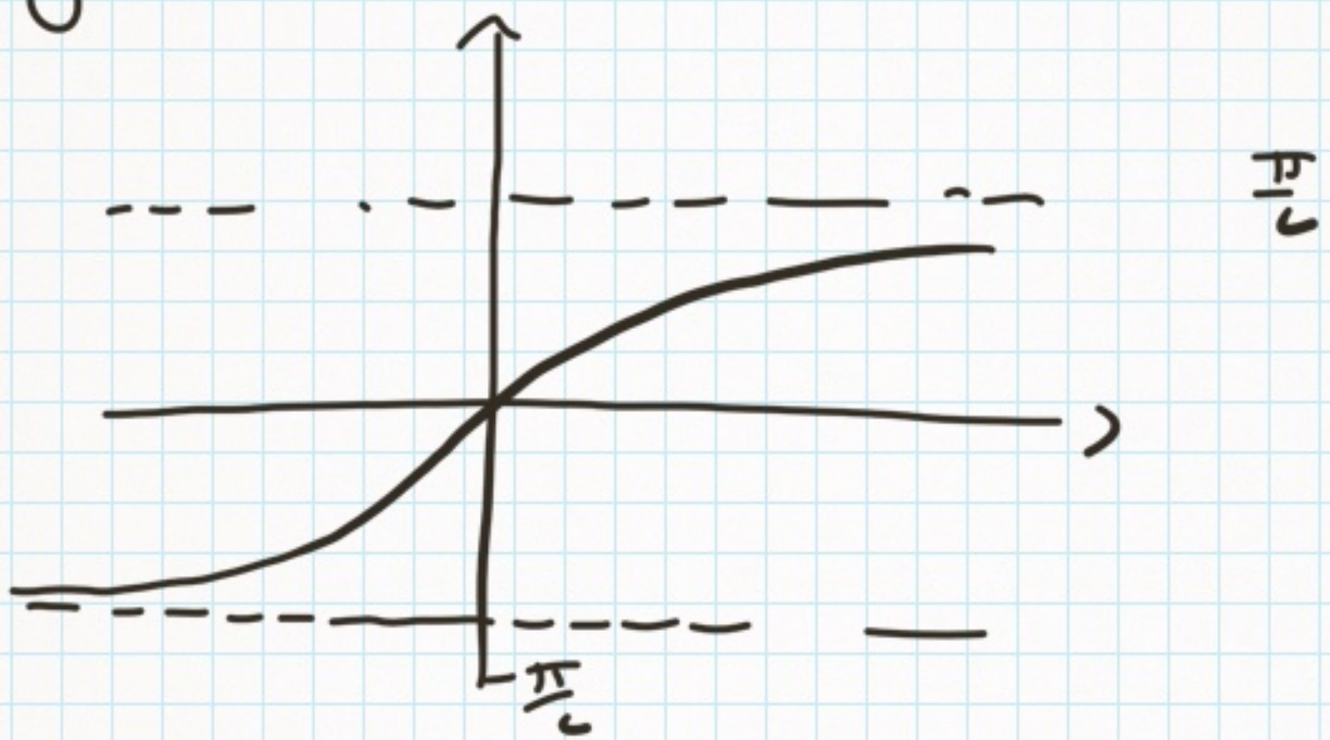


$\forall x_0 \in]-1, 1[\quad \lim_{x \to x_0} \arccos x = \arccos x_0$

$\lim_{x \to -1^+} \arccos x = \pi$

$\lim_{x \to 1^-} \arccos x = 0$

(14) $f(x) = \operatorname{arctg} x$ $\quad x \in \mathbb{R}$



$$\lim_{x \to +\infty} \operatorname{arctg} x = \frac{\pi}{2}$$

$$\lim_{x \to -\infty} \operatorname{arctg} x = -\frac{\pi}{2}$$

$$\forall x_0 \in \mathbb{R} \quad \lim_{x \to x_0} \operatorname{arctg} x = \operatorname{arctg} x_0$$

TEOREMA

Siano $f, g : I \setminus \{x_0\} \longrightarrow \mathbb{R}$

Supponiamo che

$$\lim_{x \to x_0} f(x) = l_1, \quad \lim_{x \to x_0} g(x) = l_2$$

Allora

(1) $\lim_{x \to x_0} f(x) + g(x) = l_1 + l_2$.

(2) $\lim_{x \to x_0} f(x) \cdot g(x) = l_1 \cdot l_2$

(3) $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{l_1}{l_2}$ (se ha senso il secondo)

TRANNE CHE PER LE FORME INDETERMINATE

$\left[\frac{0}{0}\right]$, $\left[\frac{\pm\infty}{\pm\infty}\right]$, $[0 \cdot (\pm\infty)]$, 0^0, 1^∞

Ludi volgono le GERARCHIE di INFINITI viste per le SUCCESSIONI e x → ±∞

$$\lim_{x\to+\infty} \frac{x^3 + x^2}{\log_3 x} = \lim_{x\to+\infty} \frac{x^3}{\log_3 x} = +\infty$$

perché $f(x) = \log_3 x$ va a $+\infty$ più lentamente di $g(x) = x^3$)

$$\lim_{x\to+\infty} \frac{3x^2 + \log x - 3}{\sqrt{2x^4+1}} = \lim_{x\to+\infty} \frac{3x^2}{\sqrt{2x^4}} =$$

$$= \lim_{x\to+\infty} \frac{3x^2}{\sqrt{2}\,x^2} = \frac{3}{\sqrt{2}}$$

$$\lim_{x\to+\infty} \sqrt{x^2+x-1} - x$$

DIFFERENZA DI INFINITI EQUIVALENTI

⇓

RAZIONALIZZARE !!

$$= \lim_{x\to+\infty} \frac{\left(\sqrt{x^2+x-1} - x\right)\left(\sqrt{x^2+x-1} + x\right)}{\sqrt{x^2+x-1} + x}$$

$$= \lim_{x\to+\infty} \frac{x^2+x-1-x^2}{\sqrt{x^2}+x} = \lim_{x\to+\infty} \frac{x}{|x|+x} = \lim_{x\to+\infty} \frac{x}{x+x} = \frac{1}{2}$$

$|x| = x$ se $x>0$

$$\cdot \lim_{x \to -\infty} \sqrt{x^2+x+1} + x \qquad [+\infty - \infty]$$

$$(a^2 - b^2) = (a+b)(a-b)$$

$$= \lim_{x \to -\infty} \frac{\left(\sqrt{x^2+x+1} + x\right)\left(\sqrt{x^2+x+1} - x\right)}{\sqrt{x^2+x+1} - x}$$

$$= \lim_{x \to -\infty} \frac{\left(\sqrt{x^2+x+1}\right)^2 - x^2}{\sqrt{x^2} - x} = \lim_{x \to -\infty} \frac{x+x+1-x}{|x| - x} =$$

$$|x| = -x \quad \text{se} \quad x < 0$$

$$= \lim_{x\to-\infty} \frac{x}{-x-x} = \lim_{x\to-\infty} \frac{x}{-2x} = -\frac{1}{2}$$

- $\lim_{x\to-\infty} \sqrt{x^2-7x+1} + x = [+\infty - \infty]$

RAZIONALIZZA 2 OLTE

$$= \lim_{x\to-\infty} \frac{(\sqrt{x^2-7x+1} + x)(\sqrt{x^2-7x+1} - x)}{\sqrt{x^2-7x+1} - x}$$

$$= \lim_{x\to-\infty} \frac{x^2-7x+1-x^2}{\sqrt{x^2}-x} = \lim_{x\to-\infty} \frac{-7x+1}{|x|-x}$$

$|x| = -x$ per $x<0$

$$= \lim_{x\to-\infty} \frac{-7x}{-x-x} = \lim_{x\to-\infty} \frac{-7x}{-2x} = \frac{7}{2}$$